

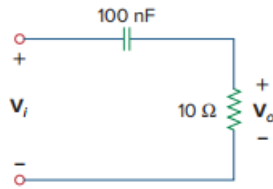
Problem

Refer to the RC circuit in Fig. 9.81.

(a) Calculate the phase shift at 2 MHz.

(b) Find the frequency where the phase shift is 45° .

Figure 9.81:



Step-by-step solution

Step 1 of 3

Refer to Figure 9.81 in the text book.

(a)

Consider the expression for the reactance due to capacitance.

$$X_c = \frac{-1}{\omega C}$$

Substitute $2\pi f$ for ω .

$$X_c = \frac{-1}{2\pi fC}$$

Substitute 2 MHz for f and 100 nF for C .

$$\begin{aligned} X_c &= \frac{-1}{2\pi(2\text{ M})(100\text{ n})} \\ &= \frac{-1}{1.2566} \\ &= -0.7958\ \Omega \end{aligned}$$

Therefore, the value of the reactance X_c is, $-0.7958\ \Omega$.

Step 2 of 3

Calculate the value of the phase shift.

$$\theta = \tan^{-1}\left(\frac{X_c}{R}\right) \dots\dots (1)$$

Substitute $-0.7958\ \Omega$ for X_c and $10\ \Omega$ for R .

$$\begin{aligned} \theta &= \tan^{-1}\left(\frac{-0.7958}{10}\right) \\ &= \tan^{-1}(-0.07958) \\ &= -4.55^\circ \\ &= 4.55^\circ \text{ lagging} \end{aligned}$$

Lagging since phase shift is negative.

Therefore, the value of the phase shift is, 4.55° lagging.

(b)

Recall equation (1).

$$\theta = \tan^{-1} \left(\frac{X_C}{R} \right)$$

Substitute $\frac{-1}{\omega C}$ for X_C .

$$\theta = \tan^{-1} \left(\frac{-1}{\omega CR} \right)$$

Substitute $2\pi f$ for ω .

$$\theta = \tan^{-1} \left(\frac{-1}{2\pi fCR} \right)$$

Treat the phase shift as lagging, $\frac{-1}{2\pi fCR}$ can be taken as $\frac{1}{2\pi fCR}$.

$$\theta = \tan^{-1} \left(\frac{1}{2\pi fCR} \right)$$

Substitute 45° for θ , 100 nF for C and 10Ω for R .

$$45^\circ = \tan^{-1} \left(\frac{1}{2\pi f(100 \text{ n})(10)} \right)$$

$$\tan 45^\circ = \frac{1}{f(6.2832 \mu)}$$

$$1 = \frac{1}{f(6.2832 \mu)}$$

$$f = \frac{1}{6.2832 \mu}$$

$$f = 159.16 \text{ kHz}$$

Therefore, the value of the frequency f is, 159.16 kHz.